EXPERIMENT 15

Report on Career Pathways and the Future of Low-Code Development

Project context: Patient Appointment Scheduling System | Zoho Creator

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Abstract

This report explores career pathways created by low-code technologies, the technical and professional skills they require, and the future of low-code development. The discussion is connected to the Patient Appointment Scheduling System developed in Zoho Creator, which uses forms, reports, dashboard components and a Deluge workflow. It explains relevant roles, skills, learning routes and future trends, including AI-assisted building, hybrid low-code/pro-code development, governance and integration.

1. Introduction

Low-code platforms help users build applications through visual tools, forms, reports and workflows, while still requiring problem solving, data modelling, process design, testing and integration skills. The Patient Appointment Scheduling System is a practical example: the appointment form captures patient and booking information, reports organize saved records, dashboard components summarize information, and a Deluge workflow supports fee calculation.

2. Career Pathways

Career role	Responsibility related to this project	Key skills to build
Low-Code Developer	Builds and maintains forms, reports, dashboard pages and workflows.	Zoho Creator, Deluge, data modelling, UI design, testing
Citizen Developer	Creates small departmental tools for clinic staff.	Domain knowledge, basic logic, platform fundamentals
Business Analyst	Maps the appointment booking process and clinic requirements.	Requirement gathering, process modelling, communication
Integration Developer	Connects appointment records to external systems and services.	REST APIs, JSON, OAuth, middleware
Solution Architect	Plans a secure, scalable design combining low-code and pro-code.	System design, security, scalability, governance
Platform Administrator	Manages users, roles, permissions and release environments.	Role-based access control, audit, deployment management
Pro-Code Extension Developer	Adds custom functions or serverless services where needed.	Python, Java or Node.js, APIs, serverless platforms
AI and Automation Specialist	Adds analytics, prediction or intelligent workflow features.	ML fundamentals, analytics tools, prompt design

3. Skills to Build

Technical Skills	Professional Skills
• Relational data modelling • Workflow scripting with Deluge • REST APIs and integration • Security and role-based access • Cloud and DevOps basics • AI and analytics fundamentals	• Requirement analysis and problem solving • Communication with stakeholders • Process thinking and documentation • Project and time management • Continuous learning

Technical depth gives credibility as an application becomes complex, while professional skills help ensure the application fits users' real needs.

4. Certification and Learning Routes

• Follow vendor learning paths and certifications for Zoho Creator where available.

• Explore transferable low-code platforms such as Microsoft Power Platform, OutSystems, Mendix and Appian.

• Build a portfolio project: document the Patient Appointment Scheduling System, its forms, reports, dashboard and workflows, and publish non-sensitive documentation or code samples on GitHub.

• Participate in hackathons, internships and developer communities.

5. Future of Low-Code Development

• AI-native building: describing an application in plain language and refining it through conversation will become more common.

• Low-code and pro-code convergence: developers will use visual tools for speed and add code where more control is needed.

• Stronger governance: organizations will manage permissions, audit, deployment and compliance more carefully.

• Wider adoption: reusable templates will support industries such as healthcare, education and hospitality.

• Role evolution: as routine form and report creation becomes more automated, value will shift toward architecture, integration, data and AI skills.

6. Relevance to an AI and ML Student

For an AI and ML student, low-code offers a quick way to turn models into usable products. A trained model can be exposed through an API and connected to a low-code front end that already provides forms, user management and dashboards. For this project, a future extension could connect a validated appointment analytics or patient no-show prediction model through an API, while Zoho Creator remains the staff-facing application. This is a proposed extension, not a feature currently shown in the screenshots.

7. Project Screenshots and Observations

The following screenshots are from the existing Zoho Creator project. They show current application components and are used as practical evidence; they do not show the proposed AI/ML extension as implemented.

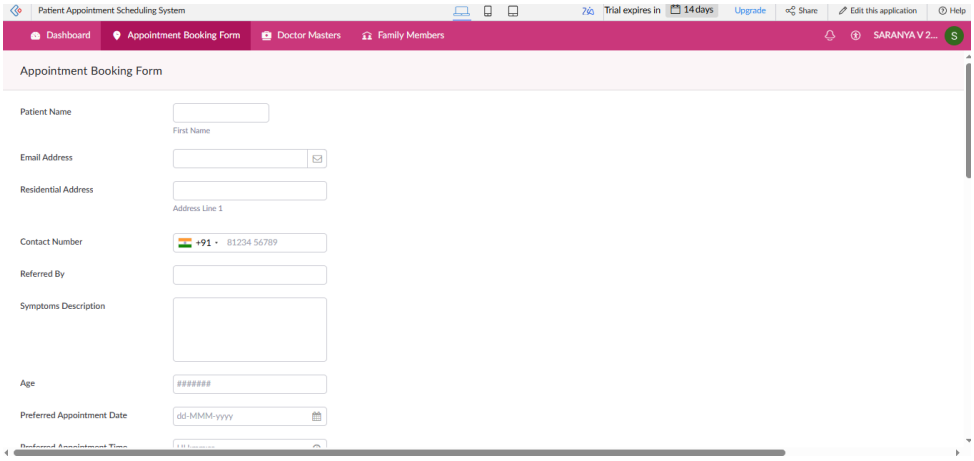


Figure 1: Appointment Booking Form of the Patient Appointment Scheduling System.

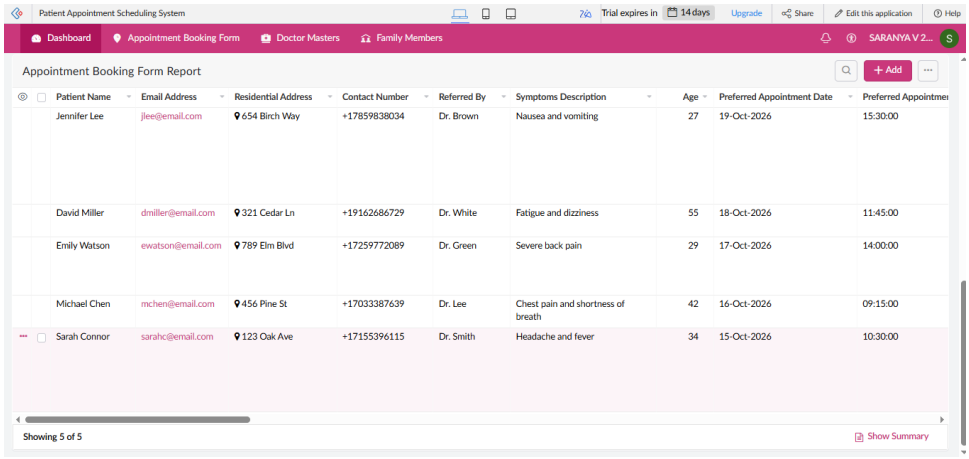


Figure 2: Appointment Booking Form Report displaying the project's sample records.

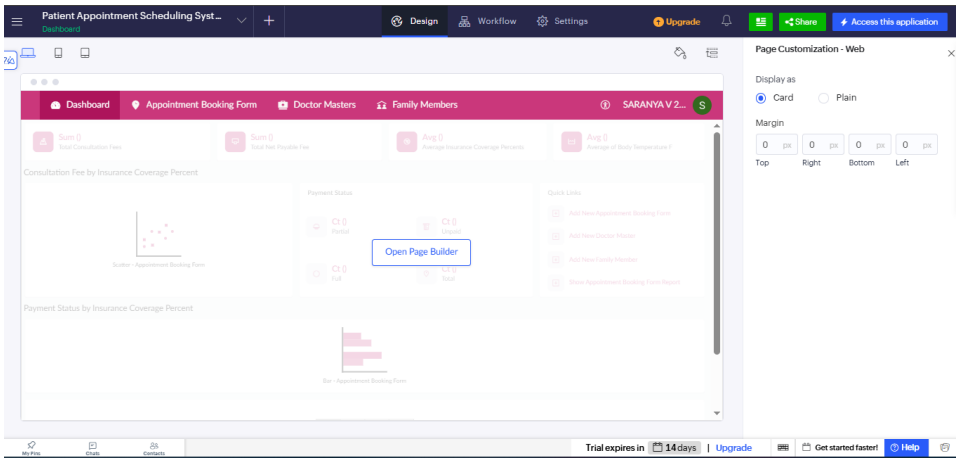


Figure 3: Dashboard layout with summary cards, payment-status section, quick links and charts.

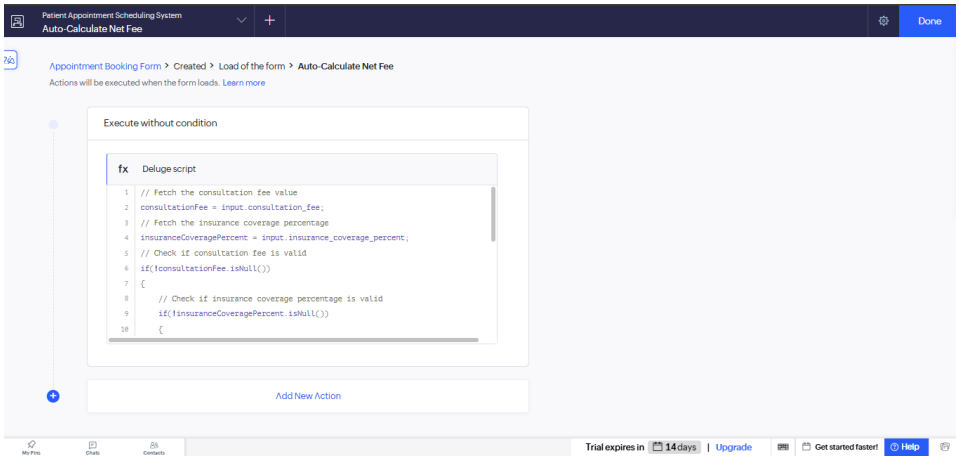


Figure 4: Existing Auto-Calculate Net Fee Deluge workflow.

8. Conclusion

Low-code development offers career opportunities that combine technical ability with business understanding. The Patient Appointment Scheduling System demonstrates practical low-code skills through forms, reports, dashboards and Deluge workflows. As AI and automation improve, long-term value will depend on strong data modelling, integrations, security, architecture and intelligent automation. A documented working portfolio helps demonstrate these skills.

9. Summary

This report covers eight career roles, technical and professional skills, learning routes, future trends and the relevance of low-code to AI/ML. The project examples connect the discussion to a realistic healthcare workflow.

10. Learning Outcome

Learnt to identify career options around low-code development, map the skills needed for each role, plan learning and portfolio activities, and explain how AI, integration and governance may shape the future of low-code.

11. Presentation Outline

• Introduction to low-code and why it is growing

• Key roles and career pathways

• Technical and professional skills

• Certification and portfolio roadmap

• Future trends: AI, governance and hybrid development

• Relevance to the Patient Appointment Scheduling System

• Conclusion and questions